UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
SEM I - 2024/2025

SECP2523 – DATABASE
ASSIGNMENT 1 – NORMALIZATION

QUESTION 1

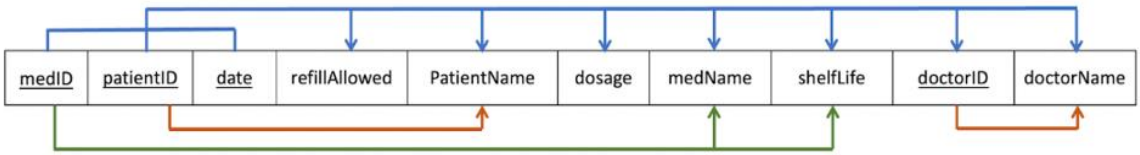


Figure 1: Patient Relation

a) Identify **ALL** functional dependencies in the Patient relation (in Figure 1).

b) Using the partial dependencies (if they exist in Figure 1), produce a set of relations in Second Normal Form (2NF).

c) Using the transitive dependencies (if they exist in Figure 1), produce a set of relations in Third Normal Form (3NF).

QUESTION 2

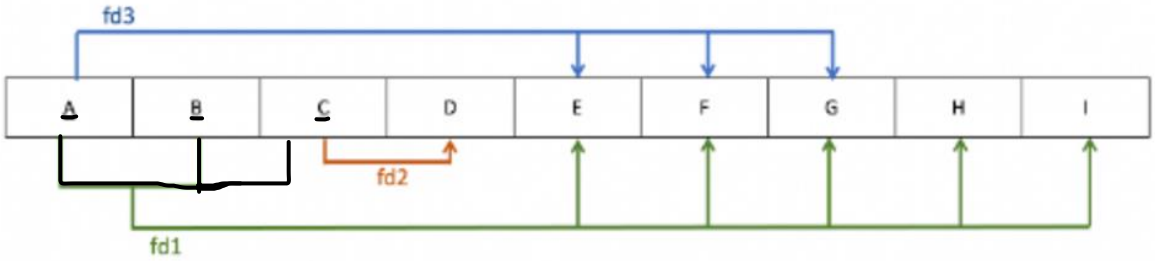


Figure 2: ABC Relation

a) Identify **ALL** functional dependencies in the ABC relation (in Figure 2).

b) Using the partial dependencies (if they exist in Figure 2), produce a set of relations in Second Normal Form (2NF).

c) Using the transitive dependencies (if they exist in Figure 2), produce a set of relations in Third Normal Form (3NF).

QUESTION 3

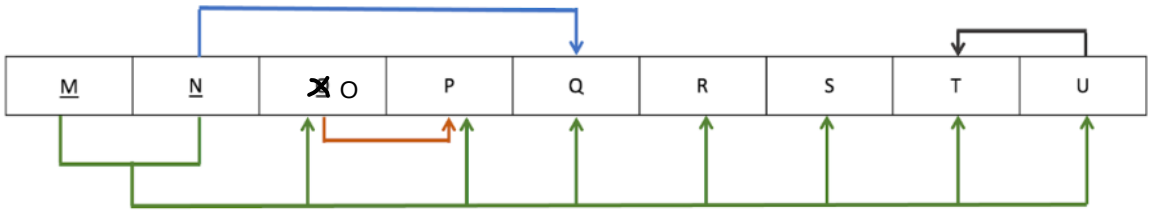


Figure 3: MNQ Relation

a) Identify **ALL** functional dependencies in the MNQ relation (in Figure 3).

b) Using the partial dependencies (if they exist in Figure 3), produce a set of relations in Second Normal Form (2NF).

c) Using the transitive dependencies (if they exist in Figure 3), produce a set of relations in Third Normal Form (3NF).

QUESTION 4

Consider the relation schema LOTS shown in **Table 3**, which describes parcels of land for sale in various counties of a state. Suppose that there are two candidate keys: PropertyID and {County, LotNo}; that is, lot numbers are unique only within each county, but PropertyID numbers are unique across counties for the entire state. PropertyID is chosen as the primary key, and it is underlined in **Table 3**.

Table 3: LOTS relation

PropertyID	**County**	**LotNo**	**Area (Acre)**	**Price (USD)**	**TaxRate**
P00121	Shawnee	LN02	0.5	500,000	5.5
P05155	Lyon	LN02	1.2	800,000	7.5
P01215	Shawnee	LN21	0.5	500,000	5.5
P20110	Lyon	LN31	1.4	1,000,000	7.5
P50551	Shawnee	LN52	0.6	600,000	5.5
P11111	Shawnee	LN31	0.7	700,000	5.5
P07123	Lyon	LN25	1.4	1,000,000	7.5

a) Identify all functional dependencies in the LOTS relation.

b) Based on the general definition of normal forms, what normal form is LOTS currently in? Justify your answer.

c) Produce a set of relational schemas for the LOTS relation, normalizing it from its current normal form to Third Normal Form (3CNF).